

APPLYING EFFECTIVE ENVIRONMENTAL FORECASTS TO MANAGEMENT STRATEGY EVALUATION OF YELLOWTAIL FLOUNDER

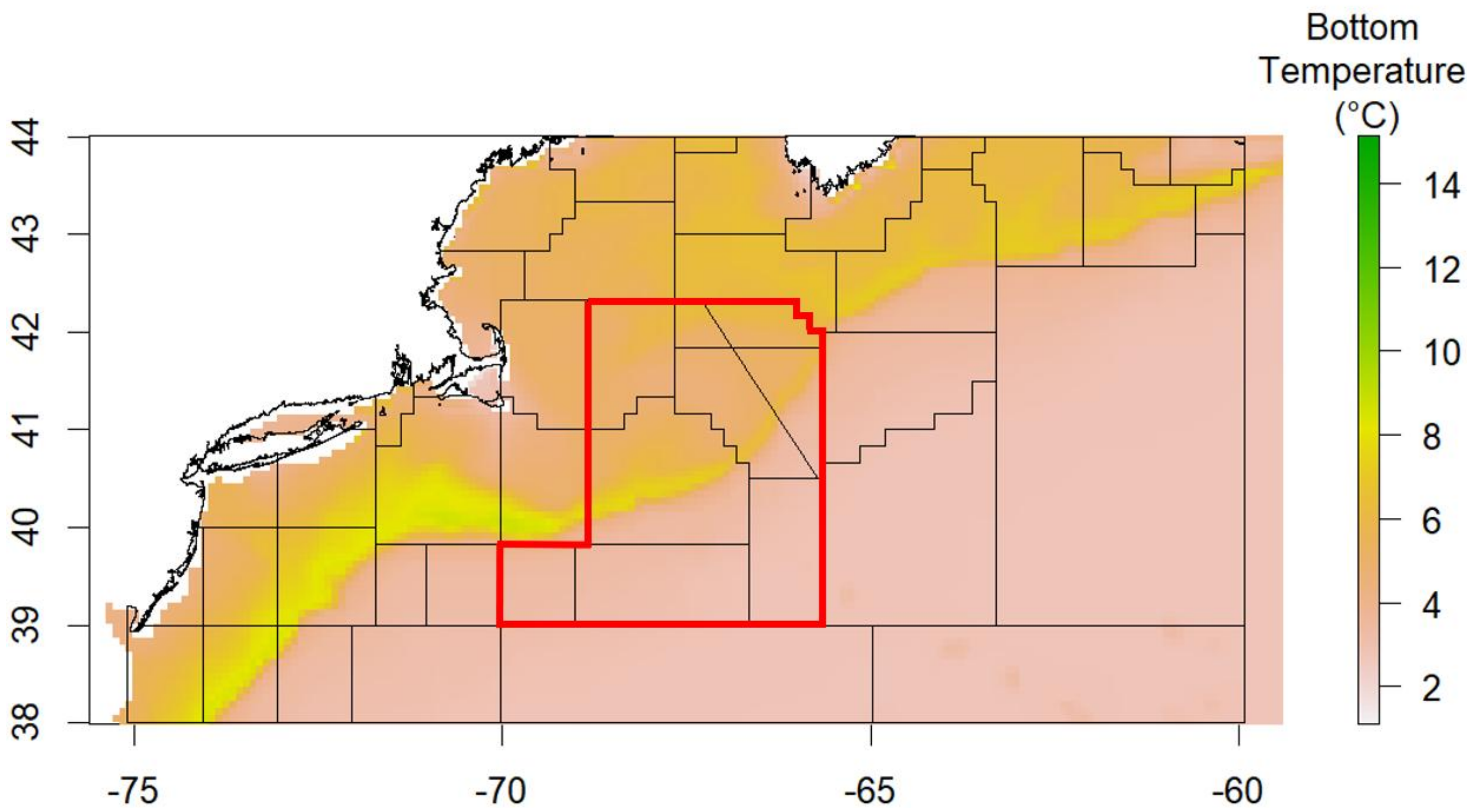
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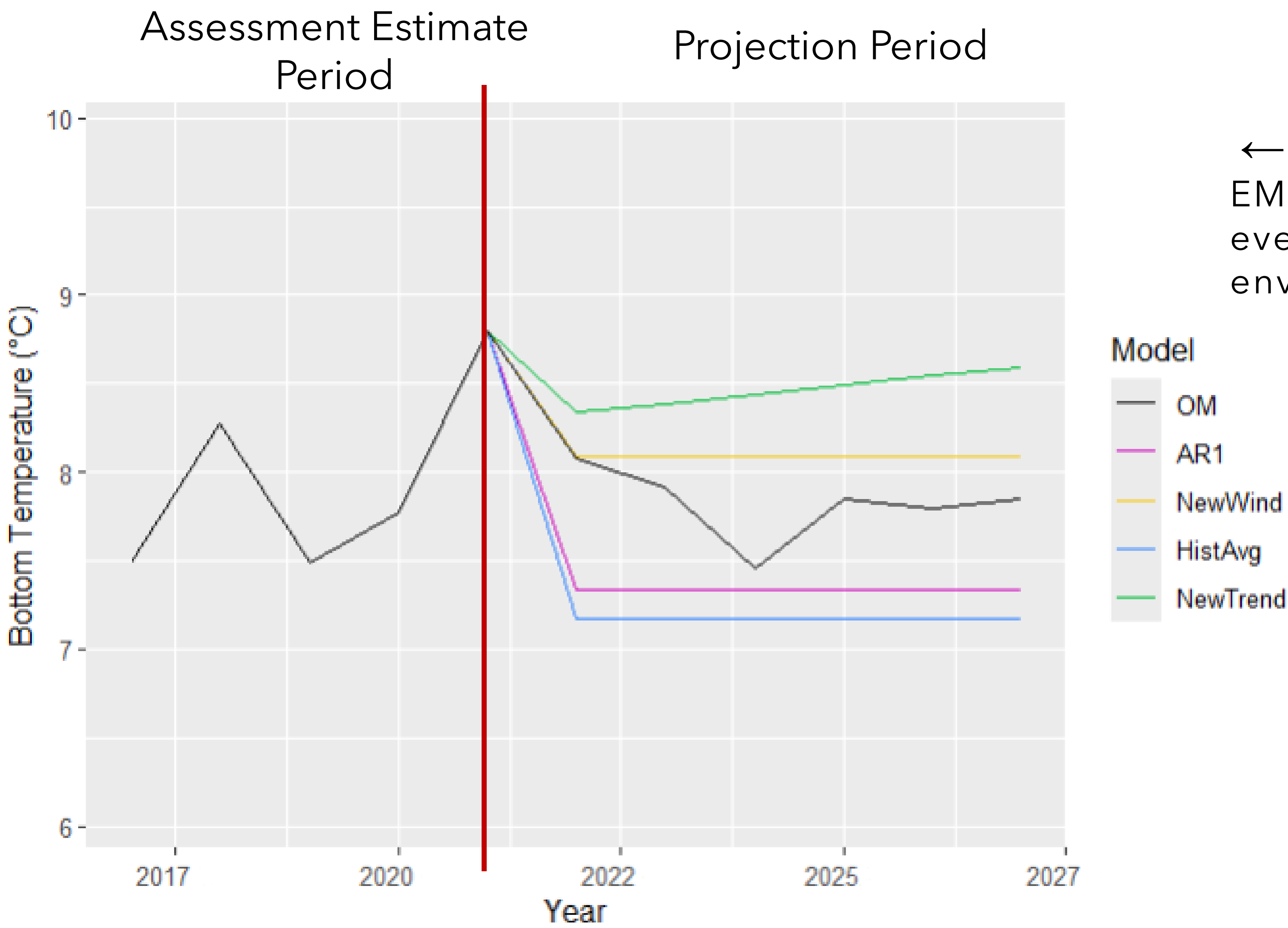


INTRODUCTION

- **Stock-specific responses to climate change creates challenges** for stock assessors and the fishing industry and a need to forecast effects of climate change on fish stocks.
- **Management Strategy Evaluation (MSE)** is a modeling framework that simulates the outcomes of management decisions.
- This study's objectives are to test how **different choices for projecting environmental covariates** within stock assessment affect long-term fishery outcomes.



↑ Figure 1: Example of bottom temperature data from MOM6. The Yellowtail Georges Bank Stock Area is outlined in red



METHODS – DATA

- Using **Georges Bank Yellowtail Flounder** (*Limanda ferruginea*) as a case study, this stock is currently overfished, and their recruitment is known to be sensitive to bottom temperature (Hyun et al 2014; Chen et al 2024).
- We will use **temperature forecasts from the Modular Ocean Model (MOM6)** (Ross et al 2021) to use as our OM environmental covariate.
- Using the **Wood's Hole Assessment Model (WHAM) and the whamMSE R package** (Stock & Miller 2021; Li 2025). We will update these packages such that new temperature data is incorporated with each MSE period, simulating periodic stock assessments.

PROJECTIONS

← Figure 3: Example temperature Projections that will be used by each EM in each Evaluation Event. The red line denotes the first evaluation event, where they models now rely on temperature projections for the environmental covariate

- **OM Projection** - Operating Model. This is the “real” condition of the fishery that will be used to simulate stock growth and ocean conditions. The estimation models will update bottom temperatures from the OM every MSE period.
- **AR1** - AutoRegressive(1) estimation process uses a linear regression of the current values of the time series on the previous value. This is an automatic setting in the WhamMSE package.
- **NewWind** - Newest Window. This projection uses the average temperature from the last five years as its projection
- **HistAvg** - Historical Average. Uses the average of the whole data series
- **NewTrend** - Newest Trend. This projection uses the linear regression of last five years of temperature data to project temperature.
- **NoTemp** - No temperature (not pictured). No Environmental covariate is used in this projection.

NEXT STEPS

- Once the whamMSE package has been properly updated, we will create a series of tradeoff plots to assess the benefits and disadvantages of the catch advice created in the Evaluations
- This research then can be expanded to additional species which have been shown to exhibit different responses beyond recruitment (e.g. growth, natural mortality) or species that respond to different environmental covariates (e.g. Cold Pool Indices, seasonal oscillations)

REFERENCES

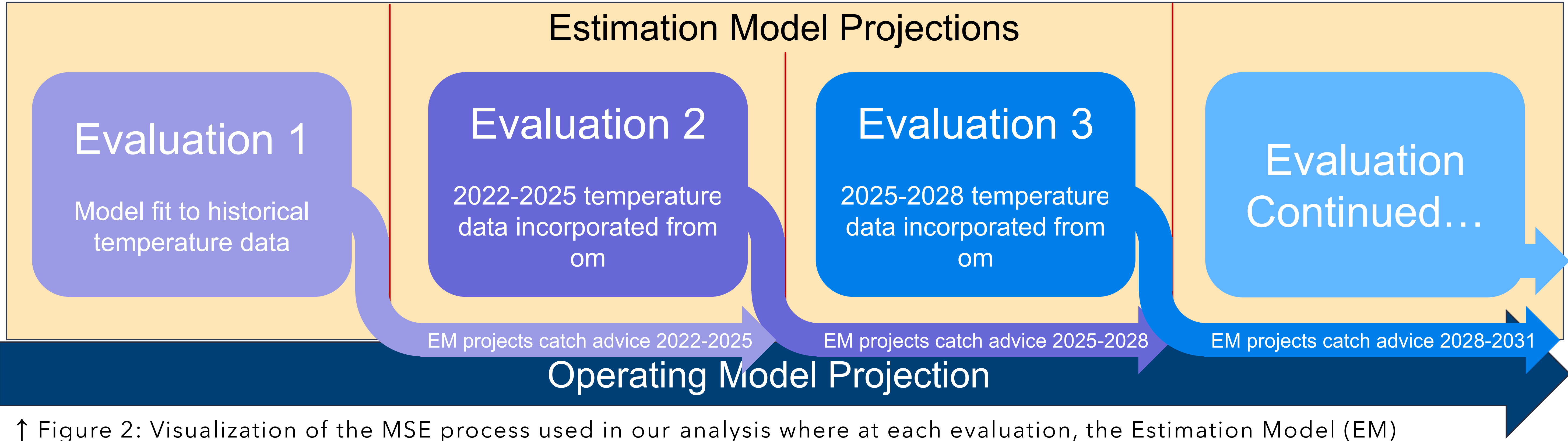
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↑ Figure 2: Visualization of the MSE process used in our analysis where at each evaluation, the Estimation Model (EM) incorporates the previous three years of temperature data into its assessment and catch advice for the next three years.